Rational Functions and Asymptotes

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Finding asymptotes and intercepts**

For $f(x) = \frac{x+2}{x-3}$, find:

a the vertical asymptote $x = 3$

b the horizontal asymptote $y = 1$

c the x-intercept $(-2, 0)$

d the y-intercept $\left(0, -\frac{2}{3}\right)$

(a) $x = 3$; (b) $y = 1$; (c) $(-2, 0)$; (d) $\left(0, -\frac{2}{3}\right)$

2 **[Foundational]** • **Target: Finding asymptotes and intercepts**

Find all asymptotes of $f(x) = \frac{2x^2+1}{x^2-4}$.

Vertical: $x = 2$ and $x = -2$. Horizontal: equal degrees, so $y = \frac{2}{1} = 2$.

3 **[Foundational]** • **Target: Finding asymptotes and intercepts**

Consider $f(x) = \frac{x^2-9}{x-3}$.

a Simplify f and identify the feature at $x = 3$. $f(x) = x + 3$ for $x \neq 3$: a hole at $(3, 6)$.

b Does the graph have a vertical asymptote? No — the discontinuity is removable, not an asymptote.

(a) $f(x) = x + 3$ for $x \neq 3$: a hole at $(3, 6)$.; (b) No — the discontinuity is removable, not an asymptote.

4 **[Foundational]** • **Target: Finding asymptotes and intercepts**

Find the slant (oblique) asymptote of $f(x) = \frac{x^2+1}{x-1}$.

Dividing: $f(x) = x + 1 + \frac{2}{x-1}$, so the slant asymptote is $y = x + 1$.

5 **[Foundational]** • **Target: Behaviour near asymptotes**

For $f(x) = \frac{x+2}{x-3}$, describe the behaviour of $f(x)$ as:

a $x \to 3^+$ $f(x) \to +\infty$

b $x \to 3^-$ $f(x) \to -\infty$

c $x \to \infty$ $f(x) \to 1$

(a) $f(x) \to +\infty$; (b) $f(x) \to -\infty$; (c) $f(x) \to 1$

6 **[Foundational]** • **Target: Behaviour near asymptotes**

Write a rational function with vertical asymptotes at $x = 1$ and $x = -4$ and horizontal asymptote $y = 0$.

One example: $f(x) = \frac{1}{(x-1)(x+4)}$ (any numerator of degree less than 2 works).

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7 [Foundational] • Target: Classifying end behaviour from degree

Without dividing, state the horizontal or slant asymptote (or neither) for each function based on the degrees of numerator and denominator:

a $f(x) = \frac{3x+1}{x^2-4}$ (degree 1 over 2) Degree of numerator $<$ denominator: horizontal asymptote $y = 0$.

b $f(x) = \frac{5x^2-1}{2x^2+3}$ (degree 2 over 2) Equal degrees: horizontal asymptote $y = \frac{5}{2}$.

c $f(x) = \frac{x^3+2}{x^2-1}$ (degree 3 over 2) Numerator degree is exactly one more: slant asymptote (no horizontal asymptote).

(a) Degree of numerator $<$ denominator: horizontal asymptote $y = 0$.; (b) Equal degrees: horizontal asymptote $y = \frac{5}{2}$.; (c) Numerator degree is exactly one more: slant asymptote (no horizontal asymptote).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Classifying end behaviour from degree

Find the slant asymptote of $f(x) = \frac{2x^2-3x+1}{x+2}$.

Dividing: $2x^2 - 3x + 1 = (x+2)(2x-7) + 15$, so $f(x) = 2x - 7 + \frac{15}{x+2}$. Slant asymptote: $y = 2x - 7$.

9 [Intermediate] • Target: Holes versus asymptotes

For $f(x) = \frac{(x-2)(x+5)}{(x-2)(x-1)}$, identify the hole and the vertical asymptote.

Common factor $(x-2)$ cancels: hole at $x = 2$ (value: $f(2) = \frac{2+5}{2-1} = 7$, so hole at $(2, 7)$). Vertical asymptote from the remaining denominator: $x = 1$.

10 [Intermediate] • Target: Holes versus asymptotes

Construct a rational function that has a hole at $x = 3$ and a vertical asymptote at $x = -1$.

One example: $f(x) = \frac{(x-3)(x+2)}{(x-3)(x+1)}$, which simplifies to $\frac{x+2}{x+1}$ with a hole at $x = 3$ and vertical asymptote at $x = -1$.

11 [Intermediate] • Target: Reading asymptotic behavior from a graph

The graph of $f(x) = \frac{1}{x-1}$ is shown. Use it to describe the behavior near $x = 1$ and as $x \to \pm\infty$.

As $x \to 1^-$, $f(x) \to -\infty$; as $x \to 1^+$, $f(x) \to +\infty$ — a vertical asymptote at $x = 1$. As $x \to \pm\infty$, $f(x) \to 0$ — a horizontal asymptote at $y = 0$.

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12 [Intermediate] • Target: Domain restrictions and sign analysis

Find the domain of $f(x) = \frac{x-1}{x^2-x-6}$, and determine the sign of $f(x)$ on the interval $(3, \infty)$.

Factor denominator: $x^2 - x - 6 = (x-3)(x+2)$. Domain: $x \neq 3, -2$. On $(3, \infty)$: numerator $x - 1 > 0$, denominator $(x-3)(x+2) > 0$ (both factors positive), so $f(x) > 0$ on this interval.

13 [Intermediate] • Target: Asymptotes in applied contexts

The average cost per unit of producing x items is $C(x) = \frac{500 + 3x}{x}$ (setup cost 500 riyals plus 3 riyals per unit). Find the horizontal asymptote and explain its meaning in context.

As $x \to \infty$, $C(x) \to 3$ (horizontal asymptote $y = 3$). This means as production volume grows very large, the average cost per unit approaches 3 riyals — the fixed setup cost gets spread so thin it becomes negligible.

14 [Intermediate] • Target: Asymptotes in applied contexts

A drug's concentration in the bloodstream is modeled by $C(t) = \frac{5t}{t^2+1}$ (mg/L, hours). Find the horizontal asymptote and interpret it, and find $C(0)$.

Since the denominator has higher degree, the horizontal asymptote is $y = 0$ — the concentration eventually approaches zero as the drug is fully metabolized. $C(0) = \frac{0}{1} = 0$ mg/L (none in the bloodstream initially, before administration takes effect).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Finding a rational function's equation from its features • Cross-Lesson: 3.03 Sum and Difference Formulas

Find the equation of a rational function with vertical asymptotes at $x = -2$ and $x = 3$, horizontal asymptote $y = 0$, and x-intercept at $x = 1$.

Numerator degree less than denominator (for $y = 0$), zero at $x = 1$: $f(x) = \frac{x-1}{(x+2)(x-3)}$.

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16 [Challenging] • Target: Holes versus asymptotes • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$? Show your complete algebraic work.

Final Answer: Limit at a is 3, $\lim_{x \to b^-} f(x) = -\infty$. First, we factor the numerator and the denominator of $f(x) = \frac{x^3 - 8}{x^2 - 4}$. The numerator is a difference of cubes: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$. The denominator is a difference of squares: $x^2 - 4 = (x - 2)(x + 2)$. Thus, $f(x) = \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)(x + 2)}$. A removable discontinuity occurs where a factor cancels out. Here, $(x - 2)$ cancels, so the removable discontinuity is at $x = a = 2$. To find the limit as x appro.

17 [Challenging] • Target: Reading asymptotic behavior from a graph • Cross-Lesson: 4.01 Circles and Parabolas

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

Final Answer: a = 5, b = 3. For a rational function to have a removable discontinuity at $x = c$, the numerator must have a factor of $(x - c)$ that cancels with a factor in the denominator. Given a removable discontinuity at $x = 2$, the numerator $x^2 - ax + 6$ must have $x = 2$ as a root. Thus, $2^2 - a(2) + 6 = 0$, which simplifies to $4 - 2a + 6 = 0$, so $10 - 2a = 0$, and $a = 5$. The numerator can be factored as $(x - 2)(x - 3)$ since $2 \times 3 = 6$ and $2 + 3 = 5$, so $a = 5$. For the function to have a vertica.

18 [Challenging] • Target: Domain restrictions and sign analysis • Cross-Lesson: 4.02 Ellipses

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x + 5}{x - 1}$.

Final Answer: $y = x + 6$. Divide: $x^2 + 5x + 5 = (x - 1)(x + 6) + 11$. Dropping the remainder term, the slant asymptote is $y = x + 6$.

19 [Challenging] • Target: Asymptotes in applied contexts • Cross-Lesson: 4.03 Hyperbolas

Find the slant (oblique) asymptote of $f(x) = \frac{3x^2 - 12x + 12}{x + 5}$.

Final Answer: y=3x-27. Dividing: $3x^2 - 12x + 12 \div (x + 5) = 3x - 27$ remainder 147, so the slant asymptote is $y = 3x - 27$.

20 [Challenging] • Target: Finding a rational function's equation from its features • Cross-Lesson: 5.01 Vectors: Components and Operations

Find the slant (oblique) asymptote of $f(x) = \frac{2x^2 + 12x + 18}{x - 5}$.

Final Answer: y=2x+22. Dividing: $2x^2 + 12x + 18 \div (x - 5) = 2x + 22$ remainder 128, so the slant asymptote is $y = 2x + 22$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Vertical Asymptotes and Holes

Find all vertical asymptotes and holes for $f(x) = \frac{x^2 - 4}{x^2 - x - 2}$.

a. Hole at $x = 2$, Vertical asymptote at $x = -1$ ✓

b. Vertical asymptotes at $x = 2$ and $x = -1$

c. Hole at $x = -1$, Vertical asymptote at $x = 2$

d. No holes, vertical asymptote at $x = 2$

Factor: $\frac{(x-2)(x+2)}{(x-2)(x+1)} = \frac{x+2}{x+1}$ for $x \neq 2$. The common factor $(x - 2)$ creates a hole at $x = 2$. The remaining zero of the denominator $x = -1$ gives a vertical asymptote.

22 [Foundational] • Target: Vertical Asymptotes and Holes

Find the coordinates (x, y) of the hole for $f(x) = \frac{x^2 - 9}{x - 3}$.

a. $(3, 6)$ ✓

b. $(3, 0)$

c. $(3, 3)$

d. $(-3, 0)$

For $x \neq 3$, $f(x) = x + 3$. Evaluating at $x = 3$ gives $y = 3 + 3 = 6$. The hole is at $(3, 6)$.

23 [Foundational] • Target: Vertical Asymptotes and Holes

How do you determine if a real zero of the denominator causes a hole or a vertical asymptote?

a. If the numerator also has the same zero of equal or greater multiplicity, it is a hole; otherwise it is a vertical asymptote ✓

b. Every zero of the denominator is an asymptote

c. If the degree of the numerator is larger, it is always a hole

d. Check the sign of $f(0)$

A factor that cancels completely between numerator and denominator creates a removable hole; an uncanceled denominator factor yields an infinite discontinuity (vertical asymptote).

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24 [Foundational] • Target: Horizontal Asymptotes

Find the horizontal asymptote of $f(x) = \frac{4x^2 + 1}{3x^2 - 5x + 2}$.

a. $y = 4/3$ ✓

b. $y = 0$

c. No horizontal asymptote

d. $y = 1/2$

The degrees of numerator and denominator are both 2. The horizontal asymptote is the ratio of leading coefficients: $y = 4/3$.

25 [Foundational] • Target: Horizontal Asymptotes

Find the horizontal asymptote of $g(x) = \frac{7x - 2}{x^3 + 4}$.

a. $y = 0$ ✓

b. $y = 7$

c. No horizontal asymptote

d. $y = -1/2$

The degree of the numerator (1) is strictly less than the degree of the denominator (3), so $y = 0$ is the horizontal asymptote.

26 [Foundational] • Target: Horizontal Asymptotes

Under what condition does a rational function have NO horizontal asymptote?

a. **When the degree of the numerator is strictly greater than the degree of the denominator** ✓

b. When the degree of the denominator is greater

c. When the degrees are equal

d. When there are no real roots

If $\deg(P) > \deg(Q)$, $f(x) \to \pm\infty$ as $x \to \pm\infty$, so there is no horizontal asymptote.

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27 [Foundational] • Target: Slant (Oblique) Asymptotes

When does a rational function $f(x) = \frac{P(x)}{Q(x)}$ possess a slant (oblique) asymptote?

a. When the degree of the numerator is exactly 1 greater than the degree of the denominator ✓

b. When the degree of the numerator is 2 greater

c. When the degree of the denominator is greater

d. When there are no vertical asymptotes

A slant asymptote occurs when $\deg(P) = \deg(Q) + 1$, producing a linear quotient via polynomial division.

28 [Foundational] • Target: Slant (Oblique) Asymptotes

Find the slant asymptote of $f(x) = \frac{2x^2 + 7x + 5}{x + 2}$.

a. $y = 2x + 3$ ✓

b. $y = 2x + 7$

c. $y = 2x - 1$

d. $y = x + 3$

Polynomial division: $\frac{2x^2 + 7x + 5}{x + 2} = (2x + 3) - \frac{1}{x + 2}$. As $x \to \pm\infty$, the remainder $\to 0$, giving slant asymptote $y = 2x + 3$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Slant (Oblique) Asymptotes

Find the slant asymptote of $g(x) = \frac{x^3 - 1}{x^2 + 1}$.

a. $y = x$ ✓

b. $y = x - 1$

c. $y = 1$

d. No slant asymptote

Dividing: $x^3 - 1 = x(x^2 + 1) - x - 1 \Rightarrow \frac{x^3 - 1}{x^2 + 1} = x - \frac{x + 1}{x^2 + 1}$. As $x \to \infty$, slant asymptote is $y = x$.

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30 [Intermediate] • Target: Intercepts and Asymptote Crossings

Find all x-intercepts and the y-intercept of $f(x) = \frac{x^2-1}{x^2-4}$.

a. **x-intercepts: $(\pm 1, 0)$, y-intercept: $(0, 1/4)$** ✓

b. x-intercepts: $(\pm 2, 0)$, y-intercept: $(0, -1/4)$

c. x-intercepts: $(\pm 1, 0)$, y-intercept: $(0, -1)$

d. x-intercepts: $(1, 0)$, y-intercept: $(0, 1/4)$

x-intercepts occur where numerator $= 0$: $x^2 - 1 = 0 \Rightarrow x = \pm 1$. y-intercept is $f(0) = \frac{-1}{-4} = 1/4$.

31 [Intermediate] • Target: Intercepts and Asymptote Crossings

Can a rational function cross its horizontal or slant asymptote?

a. **Yes, a function can cross its horizontal or slant asymptote at finite values of x** ✓

b. No, a graph can never touch any asymptote

c. Only at the y-intercept

d. Only if it has no vertical asymptotes

Unlike vertical asymptotes (where the function is undefined), horizontal and slant asymptotes only govern end behavior as $x \to \pm\infty$; curves can freely cross them at finite points.

32 [Intermediate] • Target: Intercepts and Asymptote Crossings

Find the point where $f(x) = \frac{x^2+3x-4}{x^2-1}$ crosses its horizontal asymptote $y = 1$.

a. $(-1, 1)$ is undefined; crossing is at $x = 1$ is undefined; crosses at $(1, 1)$ is a hole; no crossing point

b. It crosses at $x = 1$

c. It crosses at $x = 0$

d. **It does not cross its horizontal asymptote** ✓

Set $f(x) = 1$: $\frac{x^2+3x-4}{x^2-1} = 1 \Rightarrow x^2 + 3x - 4 = x^2 - 1 \Rightarrow 3x = 3 \Rightarrow x = 1$. But at $x = 1$, the function has a removable hole, so it does not cross.

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33 [Intermediate] • Target: Finding asymptotes and intercepts

Find the slant (oblique) asymptote of $f(x) = \frac{x^2+5x+1}{x-1}$.

a. $y = 6x$

b. $y = x + 6$ ✓

c. $y = x + 7$

d. No slant asymptote

Divide: $x^2 + 5x + 1 = (x - 1)(x + 6) + 7$. Dropping the remainder term, the slant asymptote is $y = x + 6$.

34 [Intermediate] • Target: Behaviour near asymptotes

Find the slant (oblique) asymptote of $f(x) = \frac{x^2+4x+1}{x-1}$.

a. $y = 5x$

b. No slant asymptote

c. $y = x + 5$ ✓

d. $y = x + 6$

Divide: $x^2 + 4x + 1 = (x - 1)(x + 5) + 6$. Dropping the remainder term, the slant asymptote is $y = x + 5$.

35 [Intermediate] • Target: Classifying end behaviour from degree

Find the slant (oblique) asymptote of $f(x) = \frac{x^2+5x-4}{x-1}$.

a. $y = x + 7$

b. $y = x + 6$ ✓

c. $y = 6x$

d. No slant asymptote

Divide: $x^2 + 5x - 4 = (x - 1)(x + 6) + 2$. Dropping the remainder term, the slant asymptote is $y = x + 6$.

36 [Intermediate] • Target: Holes versus asymptotes

Find the slant (oblique) asymptote of $f(x) = \frac{x^2+2x-1}{x-1}$.

a. $y = x + 4$

b. $y = 3x$

c. $y = x + 3$ ✓

d. No slant asymptote

Divide: $x^2 + 2x - 1 = (x - 1)(x + 3) + 2$. Dropping the remainder term, the slant asymptote is $y = x + 3$.

37 [Intermediate] • Target: Reading asymptotic behavior from a graph

Find the slant (oblique) asymptote of $f(x) = \frac{x^2+2x+3}{x-1}$.

a. $y = x + 3$ ✓

b. $y = 3x$

c. $y = x + 4$

d. No slant asymptote

Divide: $x^2 + 2x + 3 = (x - 1)(x + 3) + 6$. Dropping the remainder term, the slant asymptote is $y = x + 3$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Domain restrictions and sign analysis** • **Cross-Lesson: 4.02 Ellipses**

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 2x + 5}{x - 1}$.

a. $y = 3x$

b. $y = x + 4$

c. No slant asymptote

d. $y = x + 3$ ✓

Divide: $x^2 + 2x + 5 = (x - 1)(x + 3) + 8$. Dropping the remainder term, the slant asymptote is $y = x + 3$.

39 **[Challenging]** • **Target: Asymptotes in applied contexts** • **Cross-Lesson: 4.03 Hyperbolas**

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x + 3}{x - 1}$.

a. No slant asymptote

b. $y = 6x$

c. $y = x + 6$ ✓

d. $y = x + 7$

Divide: $x^2 + 5x + 3 = (x - 1)(x + 6) + 9$. Dropping the remainder term, the slant asymptote is $y = x + 6$.

40 **[Challenging]** • **Target: Finding a rational function's equation from its features** • **Cross-Lesson: 5.01 Vectors: Components and Operations**

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x - 2}{x - 1}$.

a. $y = x + 7$

b. $y = 6x$

c. $y = x + 6$ ✓

d. No slant asymptote

Divide: $x^2 + 5x - 2 = (x - 1)(x + 6) + 4$. Dropping the remainder term, the slant asymptote is $y = x + 6$.

41 **[Challenging]** • **Target: Vertical Asymptotes and Holes** • **Cross-Lesson: 5.02 The Dot Product**

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x + 2}{x - 1}$.

a. $y = x + 7$

b. $y = 6x$

c. $y = x + 6$ ✓

d. No slant asymptote

Divide: $x^2 + 5x + 2 = (x - 1)(x + 6) + 8$. Dropping the remainder term, the slant asymptote is $y = x + 6$.

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42 [Challenging] • Target: Horizontal Asymptotes • Cross-Lesson: 5.03 Parametric Equations

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x - 5}{x - 1}$.

a. No slant asymptote

b. $y = x + 6$ ✓

c. $y = x + 7$

d. $y = 6x$

Divide: $x^2 + 5x - 5 = (x - 1)(x + 6) + 1$. Dropping the remainder term, the slant asymptote is $y = x + 6$.

43 [Challenging] • Target: Slant (Oblique) Asymptotes • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 4x + 3}{x - 1}$.

a. $y = x + 6$

b. No slant asymptote

c. $y = 5x$

d. $y = x + 5$ ✓

Divide: $x^2 + 4x + 3 = (x - 1)(x + 5) + 8$. Dropping the remainder term, the slant asymptote is $y = x + 5$.

44 [Challenging] • Target: Intercepts and Asymptote Crossings • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 4x - 2}{x - 1}$.

a. $y = x + 5$ ✓

b. No slant asymptote

c. $y = 5x$

d. $y = x + 6$

Divide: $x^2 + 4x - 2 = (x - 1)(x + 5) + 3$. Dropping the remainder term, the slant asymptote is $y = x + 5$.

45 [Challenging] • Target: Finding asymptotes and intercepts • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 2x - 5}{x - 1}$.

a. $y = 3x$

b. No slant asymptote

c. $y = x + 4$

d. $y = x + 3$ ✓

Divide: $x^2 + 2x - 5 = (x - 1)(x + 3) - 2$. Dropping the remainder term, the slant asymptote is $y = x + 3$.

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